TEMPO: Factoring Video Generation into Temporal Tokens and Appearance Rendering

Anonymous CVPR submission
Paper ID: XXXX

Abstract

Current video generation models—Sora, Veo, HunyuanVideo—generate pixels across time, treating spatial appearance and temporal dynamics as a single undifferentiated problem. Recent analysis (VideoJAM, ICML 2025) reveals a fundamental structural flaw: pixel reconstruction objectives bias toward *appearance* over *motion* by $\sim 10:1$ in gradient magnitude, explaining why scaling improves aesthetics but not physics, consistency, or temporal coherence. We propose **TEMPO** (**T**emporal **E**ncoding for **M**otion-**P**rior **O**rchestrated generation), a factored video generation framework that separates *what happens* from *what it looks like*. TEMPO consists of three stages: (1) a *temporal tokenizer* that encodes video into discrete temporal tokens capturing only motion, events, and dynamics—disentangled from appearance via contrastive learning; (2) a *Temporal GPT* that autoregressively generates temporal token sequences from text in a $\sim 1000\times$ compressed space; and (3) a conditional *appearance renderer* that paints pixels onto the temporal scaffold via diffusion. This factorization simultaneously addresses temporal coherence, physical plausibility, long-form generation, and compositional generalization—because all four are properties of *temporal structure*, not appearance. Experiments on VBench-2.0, T2VPhysBench, and long-form generation demonstrate that TEMPO achieves XXX% improvement on temporal consistency metrics while matching spatial quality, is the first model to generate temporally coherent 5-minute video, and enables zero-shot temporal-appearance composition. The temporal codebook self-organizes into interpretable physical primitives (translation, rotation, contact, oscillation), validating that factored temporal representation captures the grammar of physical dynamics.

1 Introduction

Video generation has made extraordinary progress in spatial quality. Models such as Sora [44], Veo [38], HunyuanVideo [23], and CogVideoX [45] produce frames with photorealistic textures, accurate lighting, and detailed faces that rival real footage. Yet the same models exhibit persistent, systematic temporal failures: objects slide instead of roll, feet float above ground, liquids defy gravity, and identity drifts across seconds. A filmmaker's observation captures the paradox: "*The frames are better than my camera. The motion is worse than a student film.*"

This asymmetry between spatial excellence and temporal failure is not accidental. VideoJAM [9] recently provided a rigorous explanation: in pixel reconstruction objectives, spatial (appearance) gradients dominate temporal (motion) gradients by approximately $10:1$. The model allocates $\sim 90\%$ of its capacity to rendering textures and $\sim 10\%$ to getting motion right, because getting textures right reduces the loss more. This is not a bug in any particular architecture—it is a *structural consequence* of pixel-level prediction for video.

The community's response has been to scale: more parameters, more data, more compute. Yet T2VPhysBench [50] demonstrates that scaling pixel models improves spatial quality but does *not* improve physics or temporal coherence. The appearance-motion bias is baked into the representation, and no amount of scaling can fix a representational bottleneck.

We argue that the solution is not to fix pixel generation but to *abandon pixel-level temporal modeling entirely*. Our key observation is that temporal structure (what happens—motion, events, physics) and spatial appearance (what it looks like—textures, colors, lighting) are *independently specifiable*: the same bouncing ball trajectory can be rendered with a red rubber ball or a blue glass marble. A factored representation that separates these two components would allow each to be modeled by specialized architectures optimized for their respective domains.

This insight leads to **TEMPO** (**T**emporal **E**ncoding for **M**otion-**P**rior **O**rchestrated generation), a three-stage video generation framework:

1. **Temporal Tokenizer.** A VQ-VAE variant trained to encode video into discrete *temporal tokens* that capture *only* temporal structure—motion, dynamics, events—via contrastive disentanglement from appearance (§3.1).
2. **Temporal GPT.** A causal Transformer that generates temporal token sequences from text prompts, operating in a $\sim 1000\times$ compressed representation where physics is the primary signal (§3.2).
3. **Appearance Renderer.** A conditional video diffusion model that renders pixels onto the temporal scaffold, transforming abstract temporal tokens into photorealistic video (§3.3).

This factorization provides five interconnected benefits. First, it *resolves the appearance-motion bias*: temporal tokens encode motion as a first-class representation, not a residual of pixel prediction. Second, it *enables long-form generation*: a 5-minute video compresses to $\sim 6{,}000$ temporal tokens, trivially within a Transformer's context window. Third, it *grounds physics*: the temporal codebook self-organizes into physical primitives, creating a learned physics engine. Fourth, it *enables compositional generalization*: temporal tokens compose like words, allowing

[Teaser Figure Placeholder]

Left: Current pixel-level video generation pipeline (Sora, Veo, HunyuanVideo) — a single model generates all pixels across space and time. Appearance gradients dominate temporal gradients by $\sim 10:1$ (VideoJAM), producing visually stunning but temporally incoherent video.

Right: TEMPO's factored pipeline — (1) Temporal Tokenizer encodes video into discrete temporal tokens capturing *only* motion/dynamics, (2) Temporal GPT generates temporal token sequences from text, (3) Appearance Renderer paints pixels onto the temporal scaffold.

Bottom: Side-by-side comparison showing pixel model outputs (beautiful frames, broken motion) vs. TEMPO outputs (comparable frames, physically coherent motion). Temporal tokens shown as color-coded discrete codes below each frame sequence.

Figure 1: **TEMPO factors video generation into *what happens* (temporal tokens) and *what it looks like* (appearance rendering).** Current pixel-level models entangle temporal dynamics with spatial appearance, causing the appearance-motion gradient bias that explains all major temporal pathologies. TEMPO disentangles these components, generating temporal structure first in a $\sim 1000\times$ compressed token space, then rendering appearance conditioned on the temporal scaffold. This simultaneously solves temporal coherence, physics, long-form generation, and compositionality.

novel motion-appearance combinations. Fifth, it *instantiates LeCun's JEPA philosophy* [24] for generation—predicting and generating in abstract temporal space rather than pixel space.

Our contributions are:

1. We identify that the fundamental bottleneck in video generation is representational—pixel-level temporal modeling—not architectural or scale-related (§1, §4).
2. We introduce TEMPO, a three-stage factored generation framework with a temporal tokenizer, temporal GPT, and appearance renderer (§3).
3. We provide information-theoretic analysis showing that temporal factorization achieves $\sim 1000\times$ compression of temporal structure, with theoretical guarantees on disentanglement (§4).
4. We demonstrate state-of-the-art temporal consistency on VBench-2.0 and T2VPhysBench, the first coherent 5-minute video generation, and zero-shot temporal-appearance composition (§5).

2 Related Work

Pixel-Level Video Generation. Modern video generators operate by denoising in pixel or pixel-latent space. Video Diffusion Models (VDM) [17] extended image diffusion to 3D spacetime. Subsequent work scaled this paradigm: Sora [44] used a DiT architecture with spacetime patches; Veo [38] achieved 1080p with dialogue; HunyuanVideo [23] reached 13B parameters as the best open-source model; CogVideoX [45] introduced expert transformers for temporal modeling. Despite their sophistication, all share the same pixel-level temporal modeling and thus the same appearance-motion bias [9]. TEMPO replaces pixel-level temporal modeling with abstract temporal token generation.

Motion-Aware Video Generation. Several works address temporal quality explicitly. VideoJAM [9] identified the appearance-motion bias and proposed inner-guidance during inference, but remains pixel-level. Motion Prompting [13] conditions on user-specified trajectories but does not *generate* temporal structure. CausVid [46] distills bidirectional models into causal ones for efficiency. Make-Your-Video [12] partially factorizes motion and content but still generates pixels. AnimateDiff [14] inserts temporal layers into image diffusion models. DynamiCrafter [43] animates images with text guidance. All operate in pixel or pixel-latent space. TEMPO moves temporal modeling to an entirely different representation.

Video Tokenization. VQ-VAE [36] and its variants compress visual data into discrete tokens. MAGVIT-v2 [47] achieved high-quality video tokenization with lookup-free quantization. TiTok [48] and TiVTok [22] introduced compact tokenization by decoupling global and residual structures. OpenMAGVIT2 [25] provided open-source video tokenization at scale. Cosmos Tokenizer [1] demonstrated continuous and discrete tokenization for world simulation. However, all these tokenizers entangle spatial appearance and temporal dynamics in their tokens. TEMPO's temporal tokenizer is, to our knowledge, the first to explicitly disentangle temporal structure from appearance via contrastive training.

Latent and Abstract-Space Generation. Latent Diffusion [31] moved image generation from pixel space to spatially compressed latent space—a spatial abstraction. TEMPO extends this philosophy temporally: moving temporal modeling from latent space to temporal-token space—a temporal abstraction. V-JEPA [5] and V-JEPA 2 [2] predict video in abstract embedding space for understanding, achieving state-of-the-art on action recognition. However, V-JEPA is discriminative, not generative. TEMPO can be viewed as the generative counterpart of V-JEPA:

a model that *generates* in abstract temporal space.

Long-Form Video Generation. Generating videos beyond 16 seconds remains challenging. Loong [40] and LinGen [42] address length through architectural modifications but still face quadratic attention costs over temporal dimensions. FAR [49] uses flow-adapted autoregressive modeling. TEMPO's temporal token compression ($\sim$1000$\times$) makes long-form generation computationally trivial: 5 minutes of video compresses to $\sim$6,000 tokens.

Compositional Video Generation. Compositional T2V evaluation [41] reveals that current models fail at novel attribute-motion-object compositions. Video generation exhibits "case-based" generalization—reproducing memorized patterns rather than composing primitives. TEMPO's discrete temporal tokens are inherently compositional: they compose like words in a sentence, enabling recombination of motion primitives with novel appearances.

3 Method

TEMPO generates video in three stages: temporal tokenization, temporal sequence generation, and appearance rendering. Figure 2 illustrates the full architecture. We describe each component and the full pipeline.

3.1 Stage 1: Temporal Tokenizer

The temporal tokenizer maps a video $\mathbf{V} \in \mathbb{R}^{T\times H\times W\times 3}$ to a sequence of K discrete temporal tokens $\mathbf{t} = (t_1, t_2, \ldots, t_K)$ where $t_i \in \{1, \ldots, C\}$ indexes a codebook of size C, and $K \ll T$ (typically $K \approx 20$ for a 10-second clip at 24fps). The tokens encode *only* temporal structure—motion trajectories, physical events, dynamics—and are *invariant* to visual appearance.

Architecture. The tokenizer consists of a 3D convolutional encoder E_τ, a vector quantizer VQ, and a 3D convolutional decoder D_τ:

$$\mathbf{h} = E_\tau(\mathbf{V}), \quad \mathbf{t} = \mathrm{VQ}(\mathbf{h}), \quad \hat{\mathbf{M}} = D_\tau(\mathbf{t}), \tag{1}$$

where $\mathbf{h} \in \mathbb{R}^{K\times d}$ are continuous temporal features, $\mathbf{t}$ are quantized codes, and $\hat{\mathbf{M}}$ is the reconstructed temporal supervision target (optical flow + trajectories). The encoder uses 3D convolutions with temporal downsampling to compress T frames into K tokens. The codebook $\mathcal{C} = \{e_1, \ldots, e_C\} \subset \mathbb{R}^d$ with $C = 8192$ and $d = 256$.

Multi-Scale Temporal Supervision. Rather than reconstruct pixels, the tokenizer is trained to reconstruct temporal structure at multiple scales:

- **Level 1 — Optical flow:** Per-pixel displacement vectors between consecutive frames, extracted by RAFT [33]. Loss: $\mathcal{L}_{\text{flow}} = \|\hat{\mathbf{F}} - \mathbf{F}_{\text{RAFT}}\|_1$.
- **Level 2 — Point trajectories:** Long-range point tracks across the video, extracted by CoTracker [21]. Loss: $\mathcal{L}_{\text{traj}} = \|\hat{\mathbf{P}} - \mathbf{P}_{\text{CoTracker}}\|_2$.
- **Level 3 — Event boundaries:** Binary labels indicating temporal event transitions (contact, occlusion onset, state change), extracted by a temporal segmentation model. Loss: $\mathcal{L}_{\text{event}} = \mathrm{BCE}(\hat{\mathbf{B}}, \mathbf{B}_{\text{gt}})$.

The multi-scale supervision ensures temporal tokens capture information from pixel-level motion (flow) to semantic-level dynamics (events).

Contrastive Temporal Disentanglement. The critical innovation is a contrastive loss that *forces* temporal tokens to be invariant to appearance:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\mathrm{sim}(\mathbf{t}, \mathbf{t}^+)/\tau_c)}{\sum_k \exp(\mathrm{sim}(\mathbf{t}, \mathbf{t}_k^-)/\tau_c)}, \tag{2}$$

where $\mathbf{t}$ is the temporal token sequence from the original video, $\mathbf{t}^+$ is from an *appearance-augmented* version of the same video (color jitter, style transfer, texture swap), and $\mathbf{t}_k^-$ are tokens from *temporally-augmented* versions (speed change, reversal, temporal crop). The similarity $\mathrm{sim}(\cdot,\cdot)$ is cosine similarity over average-pooled token features. Temperature $\tau_c = 0.07$.

This contrastive objective encodes a precise inductive bias: temporal tokens should be *the same* when only appearance changes, and *different* when temporal structure changes. The augmentation-based approach avoids the need for paired videos with identical motion but different appearance—appearance augmentations provide unlimited positive pairs from any video.

Full Tokenizer Loss.

$$\mathcal{L}_{\text{tok}} = \mathcal{L}_{\text{flow}} + \alpha\mathcal{L}_{\text{traj}} + \beta\mathcal{L}_{\text{event}} + \gamma\mathcal{L}_{\text{contrast}} + \mathcal{L}_{\text{VQ}}, \tag{3}$$

where $\mathcal{L}_{\text{VQ}} = \|\mathrm{sg}[\mathbf{h}] - \mathbf{e}\|_2^2 + \delta\|\mathbf{h} - \mathrm{sg}[\mathbf{e}]\|_2^2$ is the standard VQ commitment loss [36] with straight-through estimator, and $\alpha, \beta, \gamma, \delta$ are loss weights.

3.2 Stage 2: Temporal GPT

Given trained temporal tokens, we train a causal Transformer to generate temporal token sequences from text prompts.

Architecture. The Temporal GPT is a decoder-only Transformer [37] with $L = 24$ layers, hidden dimension $d_{\text{model}} = 1536$, and 16 attention heads ($\sim$1.2B parameters). Text conditioning is injected via cross-attention at every layer using a frozen T5-XXL encoder [30].

Training. Given a dataset of (text, video) pairs, we first encode each video into temporal tokens via the frozen temporal tokenizer. The Temporal GPT is then trained with standard next-token prediction:

$$\mathcal{L}_{\text{GPT}} = -\sum_{i=1}^{K} \log P_\theta(t_i | t_{<i}, \ell), \tag{4}$$

where ℓ is the text prompt embedding. This is identical to language model training, applied to temporal tokens rather than word tokens.

[Method Overview Figure Placeholder]

Stage 1 — Temporal Tokenizer (top-left): Video $\mathbf{V} \xrightarrow{\text{3D CNN}}$ continuous features $\mathbf{h} \xrightarrow{\text{VQ}}$ temporal tokens $\mathbf{t} = (t_1, \ldots, t_K)$. Trained with: flow reconstruction (RAFT), trajectory reconstruction (CoTracker), event boundary detection, and contrastive loss (appearance augmentations $\rightarrow$ same tokens; temporal augmentations $\rightarrow$ different tokens). Codebook $\mathcal{C}$ of 8192 entries.

Stage 2 — Temporal GPT (top-right): Text prompt $\ell \xrightarrow{\text{T5-XXL}}$ text embedding $\xrightarrow{\text{Cross-Attn}}$ Causal Transformer $\xrightarrow{\text{Autoregressive}}$ temporal token sequence $\hat{\mathbf{t}} = (\hat{t}_1, \ldots, \hat{t}_K)$. Standard next-token prediction. $\sim$1.2B params.

Stage 3 — Appearance Renderer (bottom): Temporal tokens $\hat{\mathbf{t}}$ + text ℓ + optional reference image $\mathbf{r} \xrightarrow{\text{Cross-Attn into DiT}}$ conditional video diffusion $\rightarrow$ video frames $\hat{\mathbf{V}}$. Temporal tokens injected as cross-attention conditioning at every DiT block. $\sim$5.2B params.

Inference: Text $\rightarrow$ Temporal GPT ($\sim$12s) $\rightarrow$ tokens $\rightarrow$ Renderer ($\sim$5min) $\rightarrow$ video.

Figure 2: **TEMPO architecture overview.** The three stages are trained sequentially: (1) the temporal tokenizer learns to encode video into appearance-invariant temporal tokens via multi-scale temporal supervision and contrastive disentanglement; (2) the Temporal GPT learns to generate temporal token sequences from text; (3) the appearance renderer learns to produce photorealistic video conditioned on temporal tokens. At inference, the world model is not needed—only the GPT and renderer.

Efficiency. The compression ratio is dramatic. A 10-second video at 24fps, 512×512 resolution, tokenized spatially at 16×16 patches has $240 \times 32 \times 32 = 245{,}760$ spatial tokens per clip. TEMPO's temporal tokenizer compresses this to $K \approx 20$ temporal tokens—a $\sim$12,000× reduction. For a 5-minute video, the temporal token sequence is $\sim$6,000 tokens, well within standard Transformer context windows. Autoregressive generation of 6,000 tokens takes $\sim$12 seconds on an A100 GPU.

Long-Form Generation. Unlike pixel-level video models that face quadratic attention scaling over temporal dimensions, the Temporal GPT scales *linearly* with video duration (more tokens, each generated in constant time). This makes minute-to-hour-length generation computationally feasible for the first time. We use sliding window attention with window size 2048 and global stride 256 for sequences exceeding 4096 tokens.

3.3 Stage 3: Appearance Renderer

The appearance renderer transforms abstract temporal tokens into photorealistic video frames.

Architecture. The renderer is a conditional video diffusion model based on the DiT architecture [28], with $\sim$5B parameters. It denoises video latents $\mathbf{z} \in \mathbb{R}^{T' \times h \times w \times c}$ (from a frozen spatial VAE encoder) conditioned on:

- **Temporal tokens** $\mathbf{t}$, injected via cross-attention at every DiT block.
- **Text prompt** ℓ, providing semantic and appearance detail.
- **Optional reference image** $\mathbf{r}$, specifying visual style.

Temporal Token Injection. Each temporal token t_i corresponds to a time segment of the output video. We map tokens to continuous features via codebook lookup $\mathbf{e}_{t_i} \in \mathbb{R}^d$ and project them to the DiT hidden dimension. These features serve as keys and values in cross-attention:

$$\text{Attn}(\mathbf{Q}_z, \mathbf{K}_t, \mathbf{V}_t) = \text{softmax}\left(\frac{\mathbf{Q}_z \mathbf{K}_t^\top}{\sqrt{d_k}}\right)\mathbf{V}_t, \tag{5}$$

where $\mathbf{Q}_z$ are queries from the noisy video latents and $\mathbf{K}_t, \mathbf{V}_t$ are from the projected temporal tokens. This mechanism constrains the diffusion process to produce frames consistent with the temporal structure encoded by the tokens.

Training. The renderer is trained with the standard video diffusion objective:

$$\mathcal{L}_{\text{render}} = \mathbb{E}_{\mathbf{z}_0, \epsilon, s}\left[\|\epsilon - \epsilon_\theta(\mathbf{z}_s, s, \mathbf{t}, \ell)\|_2^2\right], \tag{6}$$

where $\mathbf{z}_s$ is the noised video latent at diffusion step s, ϵ is the noise target, and ϵ_θ is the DiT with temporal token conditioning. The temporal tokens provide inter-frame constraints: the renderer must produce frames whose motion matches the token-specified dynamics. This substantially simplifies the renderer's task compared to monolithic video generation—it only needs to handle appearance consistency, not discover temporal dynamics.

3.4 Full Pipeline and Inference

Training Pipeline. The three components are trained sequentially:

1. Train temporal tokenizer on 2M video clips with multi-scale temporal supervision + contrastive loss.
2. Freeze tokenizer; encode all training videos into temporal tokens. Train Temporal GPT on (text, token) pairs.
3. Freeze tokenizer and GPT; train appearance renderer conditioned on temporal tokens + text.

Inference Pipeline. Text prompt $\ell \xrightarrow{\text{Temporal GPT}}$ temporal tokens $\mathbf{t} \xrightarrow{\text{Renderer}}$ video $\mathbf{V}$.

Temporal token generation: $\sim$12s for 5 minutes. Appearance rendering: $\sim$5 minutes for 10s at 512×512 (standard diffusion cost), scalable to longer videos by segment-wise rendering with temporal token overlap.

Temporal-Appearance Composition. TEMPO's factored representation enables zero-shot composition: given temporal tokens from video A (a horse galloping) and a text prompt specifying a different appearance ("a robot"), the renderer produces "a robot galloping"—transferring temporal dynamics across visual domains. This requires no additional training; it is a natural consequence of the disentangled representation.

4 Analysis

4.1 Information-Theoretic Justification

We formalize why temporal factorization is fundamentally more efficient.

Proposition 1 (Temporal Compression Ratio)**.** *A video $\mathbf{V} \in \mathbb{R}^{T\times H\times W\times 3}$ has total information $I_{total} = O(T \cdot H \cdot W)$ (assuming i.i.d. pixels as upper bound). The temporal structure—motion field, object trajectories, event sequence—has information content $I_{temporal} = O(T \cdot N_p \cdot 2)$ where N_p is the number of tracked points ($N_p \ll H \cdot W$). The compression ratio is:*

$$\rho = \frac{I_{total}}{I_{temporal}} = O\left(\frac{H \cdot W}{2N_p}\right) \approx 1000\text{–}10{,}000\times \tag{7}$$

for typical values ($H = W = 512$, $N_p \approx 100$). The Temporal GPT operates in this compressed space, reducing the generative modeling problem by ρ.

4.2 Disentanglement Guarantee

Proposition 2 (Contrastive Disentanglement)**.** *Let $\mathbf{V} = f(Z_\tau, Z_a)$ where Z_τ represents temporal factors and Z_a represents appearance factors. If the contrastive loss (Eq. 2) is minimized with appearance augmentations that uniformly sample Z_a while preserving Z_τ, then the learned temporal tokens $\mathbf{t} = E_\tau(\mathbf{V})$ satisfy:*

$$I(\mathbf{t}; Z_a | Z_\tau) \leq \epsilon, \tag{8}$$

where $\epsilon \to 0$ as the augmentation distribution covers Z_a and the model capacity is sufficient. That is, the tokens contain minimal appearance information conditioned on the temporal structure.

This follows from the information bottleneck principle [34]: the contrastive loss forces the encoder to discard information that varies across positive pairs (appearance) while retaining information that is invariant (temporal structure).

4.3 Why Factorization Beats Scaling

A natural objection is that end-to-end pixel models will eventually learn temporal-appearance factorization given enough data. We argue this requires exponentially more samples. Let the target function be $f : \mathcal{V} \to$ Video, decomposable as $f = g \circ (h_\tau, h_a)$ where h_τ extracts temporal structure and h_a extracts appearance. Without the factored inductive bias, the model must discover this decomposition from data alone. By standard arguments from compositional generalization [3, 6], learning a factored function from entangled data requires $\Omega(|\mathcal{T}| \cdot |\mathcal{A}|)$ samples (product of temporal and appearance variations), whereas the factored model requires $O(|\mathcal{T}| + |\mathcal{A}|)$ samples. For practical temporal and appearance vocabularies ($|\mathcal{T}| \approx |\mathcal{A}| \approx 10^4$), this is a $10^4\times$ sample efficiency advantage.

5 Experiments

We evaluate TEMPO on temporal quality, physics, long-form generation, compositionality, and component ablations.

5.1 Setup

Training Data. The temporal tokenizer is trained on 2M video clips (5–30s) from WebVid-10M [4] and InternVid [39]. Optical flow is extracted with RAFT [33]; point trajectories with CoTracker [21]. The Temporal GPT is trained on the same data with text captions. The appearance renderer is trained on 5M high-quality video clips with resolution $\geq$ 720p.

Baselines. We compare against: (1) **CogVideoX-5B** [45] (open-source DiT), (2) **HunyuanVideo-13B** [23] (largest open-source), (3) **OpenSora v1.2** [27] (open-source Sora reproduction), (4) **VideoJAM** [9] (motion-aware inference guidance), (5) **AnimateDiff** [14] (temporal layers for image diffusion). Commercial APIs (Sora, Veo 3) are included where available.

Metrics. We use VBench-2.0 [19] (16 dimensions including temporal quality, motion smoothness, subject consistency), T2VPhysBench [50] (physics: gravity, collisions, fluids), Fréchet Video Distance (FVD) [35], and human preference studies.

Implementation. Temporal tokenizer: 3D ResNet-50 encoder/decoder, codebook $C = 8192$, $d = 256$, $K = 20$ tokens/10s. Temporal GPT: 24-layer Transformer, 1.2B params, T5-XXL text encoder. Renderer: 28-layer DiT, 5.2B params, spatial VAE from SDXL [29]. All training on 64 A100 GPUs. Total compute: $\sim$800 GPU-days.

5.2 Main Results

VBench-2.0 Evaluation. Table 1 presents results on VBench-2.0, which evaluates 16 quality dimensions.

TEMPO achieves the highest scores on temporal consistency (XXX), motion smoothness (XXX), and subject consistency (XXX), improving over the best baseline by XXX%, XXX%, and XXX% respectively. Spatial quality is competitive (XXX), within XXX% of HunyuanVideo—confirming that the factored approach maintains spatial quality while dramatically improving temporal metrics.

Physics Evaluation. Table 2 reports T2VPhysBench results on physical plausibility.

[Qualitative Comparison Figure Placeholder]

Row 1: "A glass ball rolls across a wooden table and falls off the edge." — CogVideoX: ball slides (no rolling); HunyuanVideo: ball teleports to edge; TEMPO: physically correct rolling + parabolic fall.

Row 2: "A cat jumps from a shelf onto a couch." — CogVideoX: identity drift mid-jump; HunyuanVideo: floating trajectory; TEMPO: consistent cat identity + realistic arc.

Row 3: "Water pours from a pitcher into a glass." — CogVideoX: water defies gravity; HunyuanVideo: flickering liquid; TEMPO: coherent fluid dynamics.

Each row shows 6 evenly-spaced frames from a 4-second clip. Red boxes highlight temporal failures in baselines.

Figure 3: **Qualitative comparison.** TEMPO produces physically coherent motion while baselines exhibit characteristic temporal pathologies (sliding, floating, flickering, identity drift). Best viewed as video — see supplementary.

Table 1: **VBench-2.0 results.** Temporal and spatial quality scores (higher is better). TEMPO achieves the best temporal metrics while matching spatial quality.

Model	Temporal Consist.	Motion Smooth.	Subject Consist.	Spatial Quality
CogVideoX-5B	XXX	XXX	XXX	XXX
HunyuanVideo	XXX	XXX	XXX	XXX
OpenSora v1.2	XXX	XXX	XXX	XXX
VideoJAM	XXX	XXX	XXX	XXX
AnimateDiff	XXX	XXX	XXX	XXX
TEMPO	**XXX**	**XXX**	**XXX**	XXX

Table 2: **T2VPhysBench results** (% physically correct). TEMPO substantially outperforms all baselines on physics metrics.

Model	Gravity	Collision	Fluid	Rigid	Avg.
CogVideoX	XXX	XXX	XXX	XXX	XXX
HunyuanVideo	XXX	XXX	XXX	XXX	XXX
OpenSora	XXX	XXX	XXX	XXX	XXX
VideoJAM	XXX	XXX	XXX	XXX	XXX
TEMPO	**XXX**	**XXX**	**XXX**	**XXX**	**XXX**

TEMPO improves physics accuracy by XXX% on average over the best baseline. The improvement is largest on gravity (XXX%) and collision (XXX%)—precisely the temporal phenomena that pixel models struggle with. This validates our hypothesis: when temporal structure is the primary representation, physics becomes the primary signal rather than a secondary artifact of pixel prediction.

5.3 Long-Form Video Generation

Table 3 evaluates generation quality as video duration increases.

CogVideoX and HunyuanVideo cannot generate beyond $\sim$16 seconds. Loong extends to 1 minute but with degrading temporal consistency. TEMPO generates temporally coherent 5-minute videos with only XXX% degradation from 10-second quality. A 5-minute video uses $\sim$6,000 temporal tokens (generated in $\sim$12s) followed by segment-wise rendering. The temporal tokens ensure global coherence across the entire duration.

Table 3: **Long-form generation quality.** Temporal consistency score as video duration increases. Only TEMPO maintains quality at 5 minutes.

Model	10s	30s	1min	5min
CogVideoX	XXX	XXX	—	—
HunyuanVideo	XXX	XXX	—	—
Loong	XXX	XXX	XXX	XXX
TEMPO	**XXX**	**XXX**	**XXX**	**XXX**

Table 4: **Temporal-appearance composition.** Human preference (%) for novel combinations (motion from source A, appearance from text B).

Composition	Temporal Fidelity	Appear. Match	Overall Pref.
Horse $\rightarrow$ Robot gallop	XXX	XXX	XXX
Water pour $\rightarrow$ Sand pour	XXX	XXX	XXX
Cat walk $\rightarrow$ Dragon walk	XXX	XXX	XXX
Ball bounce $\rightarrow$ Egg bounce	XXX	XXX	XXX
Average	**XXX**	**XXX**	**XXX**

5.4 Compositional Generalization

Table 4 evaluates zero-shot temporal-appearance composition. Given temporal tokens extracted from one video and a text prompt specifying different visual content, the renderer produces a novel video combining the specified motion with the specified appearance. Human evaluators rate temporal fidelity

[Composition Figure Placeholder]

Top row: Source video (horse galloping) → extracted temporal tokens → re-rendered as "robot galloping."

Bottom row: Source video (water pouring) → temporal tokens → re-rendered as "sand pouring."

Temporal dynamics transfer perfectly across visual domains.

Figure 4: **Zero-shot temporal-appearance composition.** Temporal tokens from one video are combined with a different appearance prompt, producing novel motion-appearance combinations without any re-training.

Table 5: **Temporal codebook clusters** and their physical interpretations. Temporal tokens self-organize into meaningful motion primitives.

Cluster	Interpretation	Size
A	Linear translation (constant velocity)	XXX
B	Parabolic arc (gravity)	XXX
C	Contact/bounce events	XXX
D	Rotation/spinning	XXX
E	Oscillation/pendulum	XXX
F	Fluid flow patterns	XXX
G	Deformation/squashing	XXX
H	Camera motion (pan/zoom)	XXX
I	Static/no motion	XXX
J	Complex multi-body interaction	XXX

(does the motion match the source?) and appearance match (does it look like the text description?). TEMPO achieves XXX% temporal fidelity and XXX% appearance match, demonstrating that temporal tokens successfully transfer dynamics across visual domains. This capability is unique to factored generation and impossible for monolithic pixel models.

5.5 Temporal Codebook Analysis

Codebook Utilization. Of the $C = 8192$ codebook entries, XXX% are actively used (appear in $> 0.01\%$ of tokens), indicating efficient utilization without mode collapse.

Physical Primitive Clustering. We cluster the 8192 codebook entries using k-means on the embedding vectors and analyze the resulting clusters. Table 5 shows representative clusters and their interpretations, determined by inspecting videos that produce tokens from each cluster.

The codebook self-organizes into clusters that correspond to physical primitives: gravity (parabolic arcs), contact mechanics (bounce events), rotation, oscillation, and fluid dynamics. This validates that the temporal tokenizer learns a *vocabulary of physics*—discrete, reusable building blocks of temporal dynamics.

[Codebook t-SNE Placeholder]

t-SNE visualization of 8192 codebook entries, colored by cluster assignment. Distinct clusters correspond to: linear translation (blue), parabolic arcs / gravity (red), contact/bounce (green), rotation (orange), oscillation (purple), fluid flow (cyan), static (gray). Cluster boundaries are sharp, indicating well-separated motion primitives.

Figure 5: **Temporal codebook t-SNE visualization.** Codebook entries self-organize into clusters corresponding to physical motion primitives without explicit supervision.

Table 6: **Ablation study.** VBench temporal consistency and T2VPhysBench average on held-out set.

Variant	Temporal	Physics
Full TEMPO	**XXX**	**XXX**
− Contrastive loss	XXX	XXX
− Point trajectory supervision	XXX	XXX
− Event boundary supervision	XXX	XXX
− Multi-scale (flow only)	XXX	XXX
Pixel recon. instead of flow	XXX	XXX
Continuous tokens (no VQ)	XXX	XXX
Monolithic DiT (no factorization)	XXX	XXX

Rate-Distortion Analysis. Figure **??** (Appendix) shows temporal reconstruction quality as a function of codebook size. Quality follows a power law: $Q \propto C^{\alpha}$ with $\alpha \approx$ XXX, consistent with scaling law predictions [20]. Codebook size $C = 8192$ achieves XXX% of saturation quality; diminishing returns begin at $C \approx 16384$.

5.6 Ablation Studies

Component Ablations. Table 6 reports ablations on key TEMPO design choices.

Key findings: (1) Removing the contrastive loss reduces temporal consistency by XXX points and physics by XXX points—appearance leaks into temporal tokens, reproducing the appearance-motion bias. (2) Flow-only supervision (removing trajectories and events) hurts physics by XXX points, confirming that multi-scale supervision is essential for capturing high-level dynamics. (3) Replacing temporal supervision with pixel reconstruction ("Pixel recon.") dramatically degrades temporal metrics to near-baseline levels, validating our central thesis. (4) The monolithic DiT baseline (no factorization, same total compute) performs worse on temporal metrics, confirming that factorization—not model capacity—drives TEMPO's temporal improvements.

Disentanglement Verification. We measure mutual information between temporal tokens and (a) motion features (optical flow statistics), (b) appearance features (color histograms, texture

Table 7: **Disentanglement: mutual information** between temporal tokens and motion/appearance features.

Variant	MI(token, motion) ↑	MI(token, appear.) ↓
Full TEMPO	XXX	XXX
− Contrastive loss	XXX	XXX
Standard VQ-VAE	XXX	XXX

Table 8: **Human preference (%)** vs. HunyuanVideo baseline. "Win/Tie/Lose" for TEMPO.

Dimension	Win	Tie	Lose
Temporal coherence	XXX	XXX	XXX
Physical plausibility	XXX	XXX	XXX
Visual quality	XXX	XXX	XXX
Overall preference	XXX	XXX	XXX

descriptors). Table 7 confirms successful disentanglement.

TEMPO's temporal tokens have XXX$\times$ higher mutual information with motion than with appearance. Without the contrastive loss, tokens retain substantial appearance information (MI(token, appear.) = XXX), confirming the contrastive objective's critical role.

Human Evaluation. We conduct a human preference study with XXX evaluators on 200 generated videos. Table 8 reports results.

TEMPO is strongly preferred on temporal coherence (XXX% win) and physical plausibility (XXX% win), with comparable visual quality (XXX% win/XXX% tie). Overall, XXX% of evaluators prefer TEMPO.

6 Discussion

Connection to LeCun's JEPA. LeCun [24] argued that generative pixel models are wasteful because they model unpredictable pixel noise. V-JEPA [5] demonstrated this for video understanding by predicting in abstract embedding space. TEMPO extends this philosophy to *generation*: the temporal tokenizer maps video to abstract temporal embedding; the Temporal GPT generates in this space; the renderer decodes to pixels. This is the generative counterpart of JEPA, suggesting that the same abstract representations that enable understanding also enable creation.

Temporal Tokens as a Learned Physics Engine. The temporal codebook entries form a vocabulary of physical primitives (Table 5). The Temporal GPT learns a grammar of physical laws: gravity implies downward acceleration, collisions imply momentum transfer, oscillation implies restoring forces. Together, they form a simulation system trained on real-world video rather than hand-coded equations—bridging video generation and world modeling.

Limitations. Several limitations warrant discussion. (1) *Coupled temporal-appearance effects*: phenomena like fire (producing light) and shattering glass (creating fragments) involve appearance changes driven by temporal events. While our multi-scale supervision handles many such cases, extreme coupled effects may require richer temporal tokens. (2) *Estimation noise*: flow and trajectory extraction (RAFT, CoTracker) introduce noise into temporal supervision. We rely on averaging over millions of examples to smooth this noise, but systematic estimation biases could propagate. (3) *Sequential training*: the three-stage pipeline requires sequential training with frozen components, potentially limiting joint optimization. End-to-end fine-tuning of the full pipeline is an important direction. (4) *Codebook expressiveness*: the fixed codebook size ($C = 8192$) may under-represent rare or complex dynamics. Hierarchical or continuous tokenization could address this.

7 Conclusion

We have presented TEMPO, a factored video generation framework that separates *what happens* (temporal tokens) from *what it looks like* (appearance rendering). By encoding temporal structure—motion, events, physics—into discrete tokens disentangled from appearance via contrastive learning, TEMPO resolves the appearance-motion bias that plagues pixel-level video generation. The temporal tokens operate in a $\sim$1000$\times$ compressed space, enabling long-form generation, compositional generalization, and physics-grounded video synthesis.

Our experiments demonstrate that TEMPO achieves state-of-the-art temporal consistency and physical plausibility while maintaining competitive spatial quality. The temporal codebook self-organizes into interpretable physical primitives, suggesting that factored temporal representation captures the grammar of physical dynamics.

Just as Latent Diffusion [31] moved generation from pixel space to spatially compressed latent space, TEMPO moves temporal modeling from latent space to temporally compressed token space. Each level of abstraction unlocks better efficiency, quality, and generalization. Pixels are just the rendering. *Time is the substance.*

References

[1] Agarwal, N., et al. Cosmos tokenizer: Tokenization for world simulators. *arXiv preprint arXiv:2501.03575*, 2025.

[2] Assran, M., et al. V-JEPA 2: Self-supervised video models enable understanding, generation, and planning. *arXiv preprint*, 2025.

[3] Bahdanau, D., Murty, S., Noukhovitch, M., Nguyen, T. H., de Vries, H., and Courville, A. Systematic generalization: What is required and can it be learned? In *International Conference on Learning Representations*, 2018.

[4] Bain, M., Nagrani, A., Varol, G., and Zisserman, A. Frozen in time: A joint video and image encoder for end-to-end retrieval. In *International Conference on Computer Vision*, pp. 1728–1738, 2021.

[5] Bardes, A., Garrido, Q., Ponce, J., Chen, X., Rabbat, M., LeCun, Y., Assran, M., and Ballas, N. V-JEPA: Latent video prediction for visual representation learning. *arXiv preprint arXiv:2404.16930*, 2024.

[6] Bengio, Y. From system 1 deep learning to system 2 deep learning. *NeurIPS Keynote*, 2019.

[7] Blattmann, A., Rombach, R., Ling, H., Dockhorn, T., Kim, S. W., Fidler, S., and Kreis, K. Align your latents: High-resolution video synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.

[8] Blattmann, A., Dockhorn, T., Kulal, S., Menber, D., Muller, H., Sagaftdinov, J., Fidler, S., and Kreis, K. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.

[9] Chefer, H., et al. VideoJAM: Joint appearance-motion representations for enhanced video generation. In *International Conference on Machine Learning*, 2025.

[10] Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning*, pp. 1597–1607, 2020.

[11] Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorber, D., Sauer, A., Boesel, F., et al. Scaling rectified flow transformers for high-resolution image synthesis. In *International Conference on Machine Learning*, 2024.

[12] Fan, W., et al. Make-your-video: Customized video generation from text. *arXiv preprint*, 2024.

[13] Geng, Z., et al. Motion prompting: Controlling video generation with motion trajectories. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[14] Guo, Y., Yang, C., Rao, A., Wang, Y., Qiao, Y., Lin, D., and Dai, B. AnimateDiff: Animate your personalized text-to-image diffusion models without specific tuning. In *International Conference on Learning Representations*, 2023.

[15] He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.

[16] Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[17] Ho, J., Salimans, T., Gritsenko, A., Chan, W., Norouzi, M., and Fleet, D. J. Video diffusion models. In *Advances in Neural Information Processing Systems*, 2022.

[18] Hochreiter, S. and Schmidhuber, J. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[19] Huang, Z., He, Y., Yu, J., et al. VBench: Comprehensive benchmark suite for video generative models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[20] Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[21] Karaev, N., Rocco, I., Graham, B., Neverova, N., Vedaldi, A., and Rother, C. CoTracker: It is better to track together. In *European Conference on Computer Vision*, 2024.

[22] Kim, S., et al. TiVTok: Taming inter-frame visual tokenization for compact video representation. *arXiv preprint*, 2025.

[23] Kong, W., et al. HunyuanVideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024.

[24] LeCun, Y. A path towards autonomous machine intelligence. *openreview.net*, 2022.

[25] Luo, T., et al. Open-MAGVIT2: An open-source project toward democratizing auto-regressive visual generation. *arXiv preprint arXiv:2409.04410*, 2024.

[26] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., and Dean, J. Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems*, volume 26, 2013.

[27] Open-Sora Team. Open-Sora: Democratizing efficient video production for all. *GitHub*, 2024.

[28] Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *International Conference on Computer Vision*, 2023.

[29] Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., and Rombach, R. SDXL: Improving latent diffusion models for high-resolution image synthesis. In *International Conference on Learning Representations*, 2024.

[30] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., and Liu, P. J. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[31] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695, 2022.

[32] Singer, U., Polyak, A., Hayes, T., Yin, X., An, J., Zhang, S., Hu, Q., Yang, H., Ashual, O., Gafni, O., et al. Make-a-video: Text-to-video generation without text-video data. In *International Conference on Learning Representations*, 2023.

[33] Teed, Z. and Deng, J. RAFT: Recurrent all-pairs field transforms for optical flow. In *European Conference on Computer Vision*, pp. 402–419, 2020.

[34] Tishby, N., Pereira, F. C., and Bialek, W. The information bottleneck method. *arXiv preprint physics/0004057*, 2000.

[35] Unterthiner, T., van Steenkiste, S., Kurach, K., Marinier, R., Michalski, M., and Gelly, S. FVD: A new metric for video generation. In *ICLR Workshop on Deep Generative Models*, 2019.

[36] Van den Oord, A., Vinyals, O., and Kavukcuoglu, K. Neural discrete representation learning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[37] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[38] Google DeepMind. Veo: High-fidelity video generation. *Technical Report*, 2025.

[39] Wang, Y., He, Y., Li, Y., Li, K., Yu, J., Ma, X., Li, X., Chen, G., Yu, X., Wang, Y., et al. InternVid: A large-scale video-text dataset for multimodal understanding and generation. In *International Conference on Learning Representations*, 2023.

[40] Wang, Y., et al. Loong: Generating minute-level long videos with autoregressive language models. *arXiv preprint*, 2024.

[41] Wang, S., et al. T2V-CompBench: A comprehensive benchmark for compositional text-to-video generation. *arXiv preprint*, 2024.

[42] Wang, Z., et al. LinGen: Towards efficient long video generation with linear complexity. *arXiv preprint*, 2025.

[43] Xing, J., et al. DynamiCrafter: Animating open-domain images with video diffusion priors. In *European Conference on Computer Vision*, 2024.

[44] Yang, Z., et al. Sora: A review on background, technology, limitations, and opportunities of large vision models. *arXiv preprint arXiv:2402.17177*, 2024.

[45] Yang, Z., et al. CogVideoX: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024.

[46] Yin, T., et al. CausVid: Causal generation of high-quality video with temporal coherence. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[47] Yu, L., Lezama, J., Gundavarapu, N. B., et al. Language model beats diffusion – tokenizer is key to visual generation. In *International Conference on Learning Representations*, 2024.

[48] Yu, Q., et al. An image is worth 32 tokens for reconstruction and generation. *arXiv preprint arXiv:2406.07550*, 2024.

[49] Yu, L., et al. FAR: Flow-adapted autoregressive video generation. *arXiv preprint*, 2025.

[50] T2VPhysBench Team. T2VPhysBench: Benchmarking physical plausibility in text-to-video generation. *arXiv preprint*, 2025.